Machine Learning and Data Mining (ST443)

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LSE

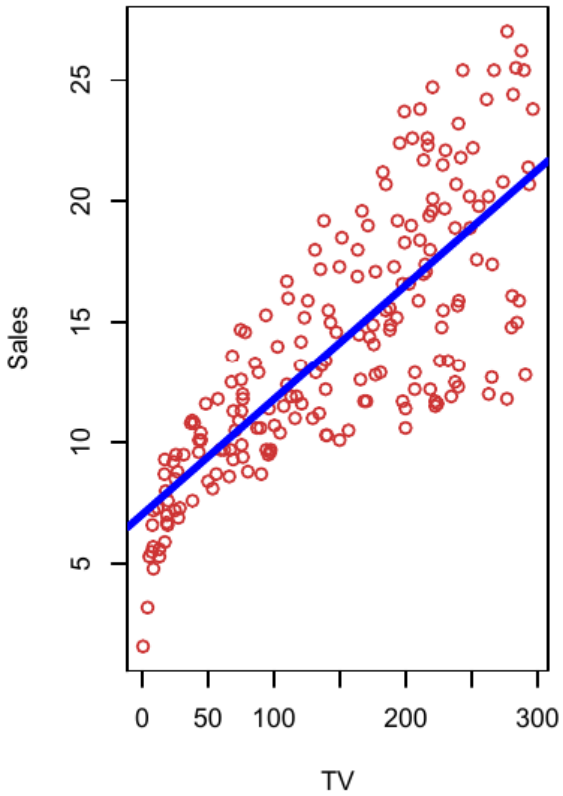
Content: statistical learning (Ch 2 @ ISLP)

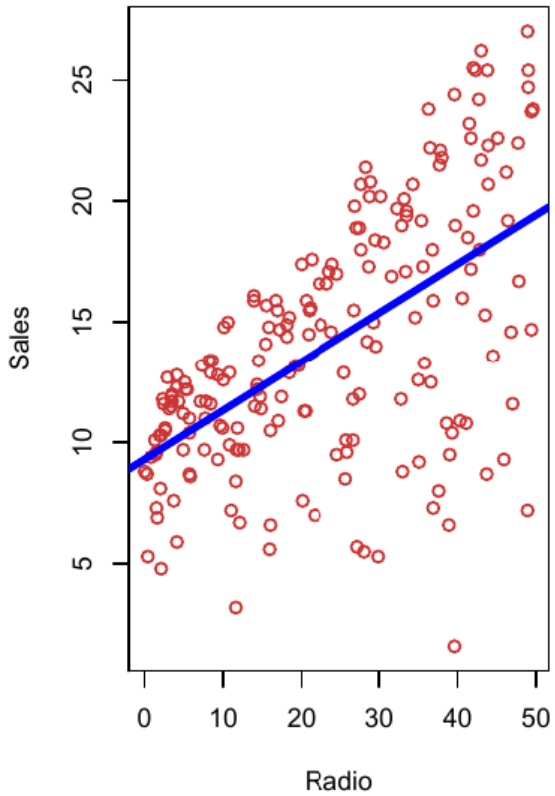
- prediction, inference,
- parametric, non-parametric methods,
- regression:
  - model accuracy, training/test MSE,
  - bias-variance tradeoff, under-/overfitting.
- classification: Bayes rate, Bayes classifier, KNN classifier.
- broader context: risk.

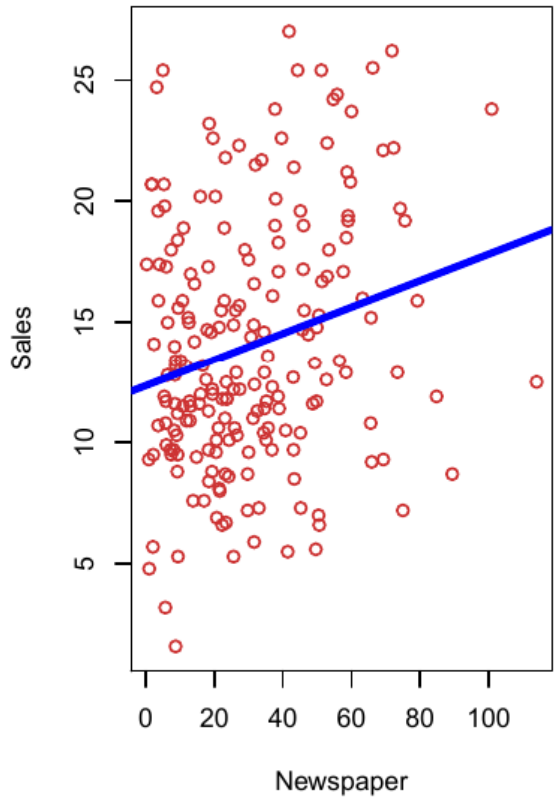
Sales example

Aim

Develop a model of sales based on advertising budget in different media (TV, radio, newspaper).







Sales example – continued

- output (sales): Y.
- input (budget): on
  - TV: X_1,
  - radio: X_2,
  - newspaper: X_3.
- nicknames:
  - output = response, dependent variable.
  - input = predictors, independent variables, features.



Relation between $Y \in \mathbb{R}$ and $X = (X_i)_{i=1}^p \in \mathbb{R}^p$

$$Y = f(X) + \epsilon. \tag{1}$$

- f: fixed, but unknown, capturing the relation of X and Y.
- ϵ: error term; $\mathbb{E}(\epsilon) = 0$, $X \perp\!\!\!\perp \epsilon$ (i.e., X and ϵ are independent).
- Given: samples $\{(\mathbf{x}_i, y_i)\}_{i \in [n]} \subset \mathbb{R}^p \times \mathbb{R}$ [realizations of (1)].
- Aim: estimate f.

Why do we estimate f?

1. prediction:
   - $\hat{f}$: blackbox; we only care about accurate prediction $[\hat{Y} = \hat{f}(X)]$.
2. inference: identifying
   1. important predictors (from a large pool of candidates).
   2. positive/negative influence of X_i-s on the output Y.
   3. nature of relation: linear/more complicated?







Inference example: sales



1. which media is mostly related to sales.
2. which media increases/decreases sales.
3. how big is the increase associated to 1 unit of TV budget boost.

Prediction & inference example: real estate

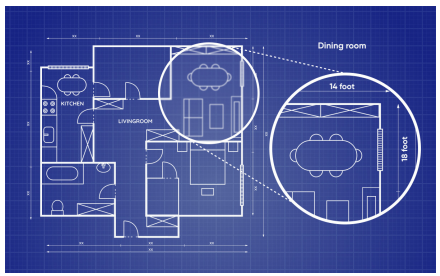


interest (note: TFL @ Capstone 2024-2025)

(i) prediction: value of home vs its characteristics (air quality, distance to river, size of the house, crime rate, . . .); (ii) inference: how much extra will a house worth with a river view



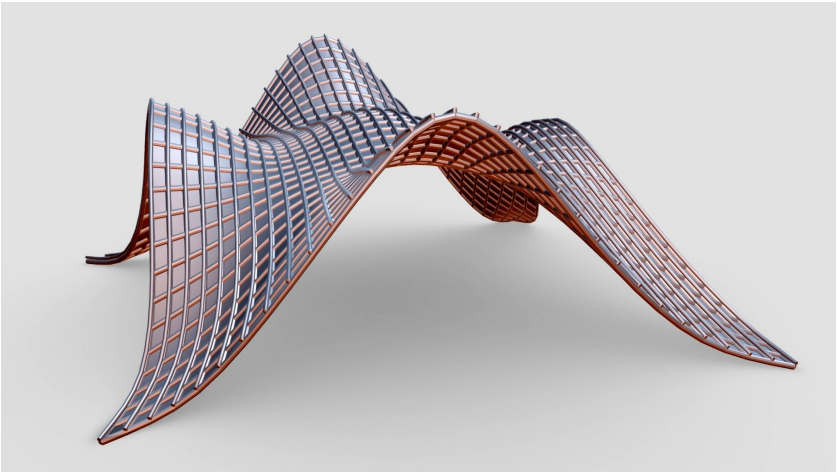






How is f estimated?

- Training data: $\{(\mathbf{x}_i, y_i)\}_{i \in [n]}$.
- Goal: find $\hat{f}$ such that $\hat{f}(X) \approx Y$ for any realization (X, Y).
- 2 main families of methods: parametric and non-parametric.



2 steps:

1. Pick a model: say linear

$$f(X) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p.$$

Estimation of f boils down to that of $\boldsymbol{\beta} = (\beta_j)_{j=0}^{p} \in \mathbb{R}^{p+1}$.

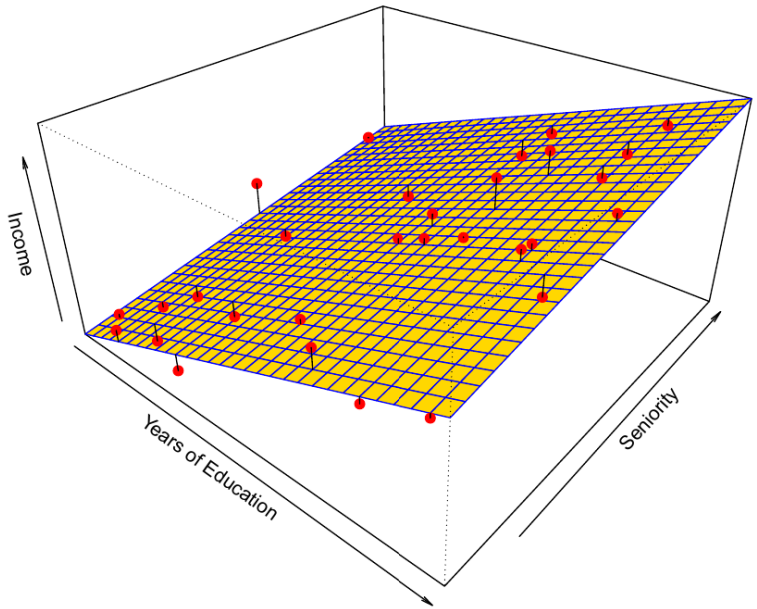
2. Fit/train the model: estimate the parameters by

$$Y \approx \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p$$

using for instance (ordinary) least squares.

Parametric methods – continued

- Pro: Estimating $\beta \in \mathbb{R}^{p+1}$ can be easier than that of a 'general' f.

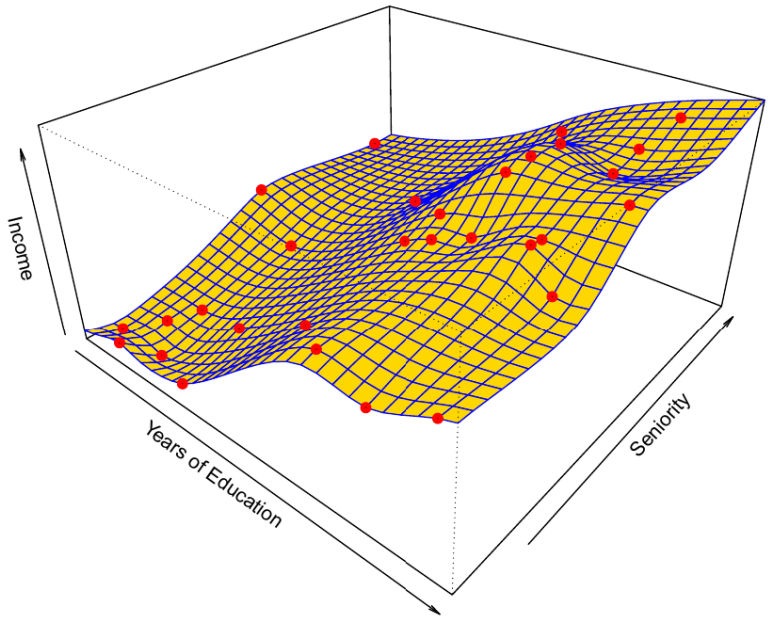


$$\text{income} \approx \beta_0 + \beta_1 \times \text{education} + \beta_2 \times \text{seniority}.$$

- Con: The picked parametric model might not match well the true f.

Parametric methods – continued

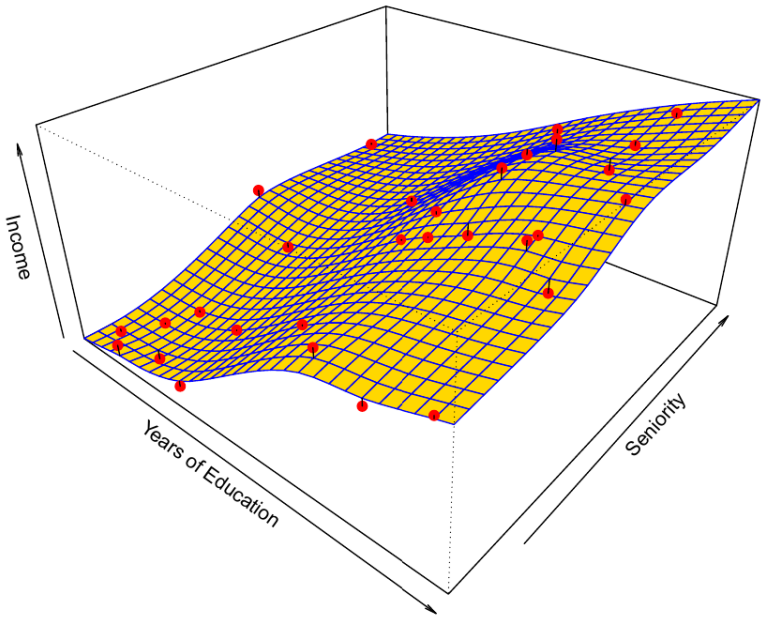
- If big mismatch $\Rightarrow$
  - more flexible models $\Rightarrow$
  - larger number of parameters $\Rightarrow$
  - potential overfitting. Ex (zero training error):



- Linear model: good baseline, especially for small sample size (n).

Nonparametric methods

- Idea: avoid parametric assumption, fit to data, while being smooth.
- Pro: can adapt to a wide range of shapes.
- Con: require significantly more samples to fit. Ex (thin-plate spline):



Their smoothness parameter has to be chosen appropriately.

Prediction and interpretability



Why choose restrictive models?

- sample efficiency.
- interpretability (see inference).

Model accuracy

No free lunch theorem

- Q: Is there a best method?
- A: No technique is superior over all datasets.

Quality of fit (regression):

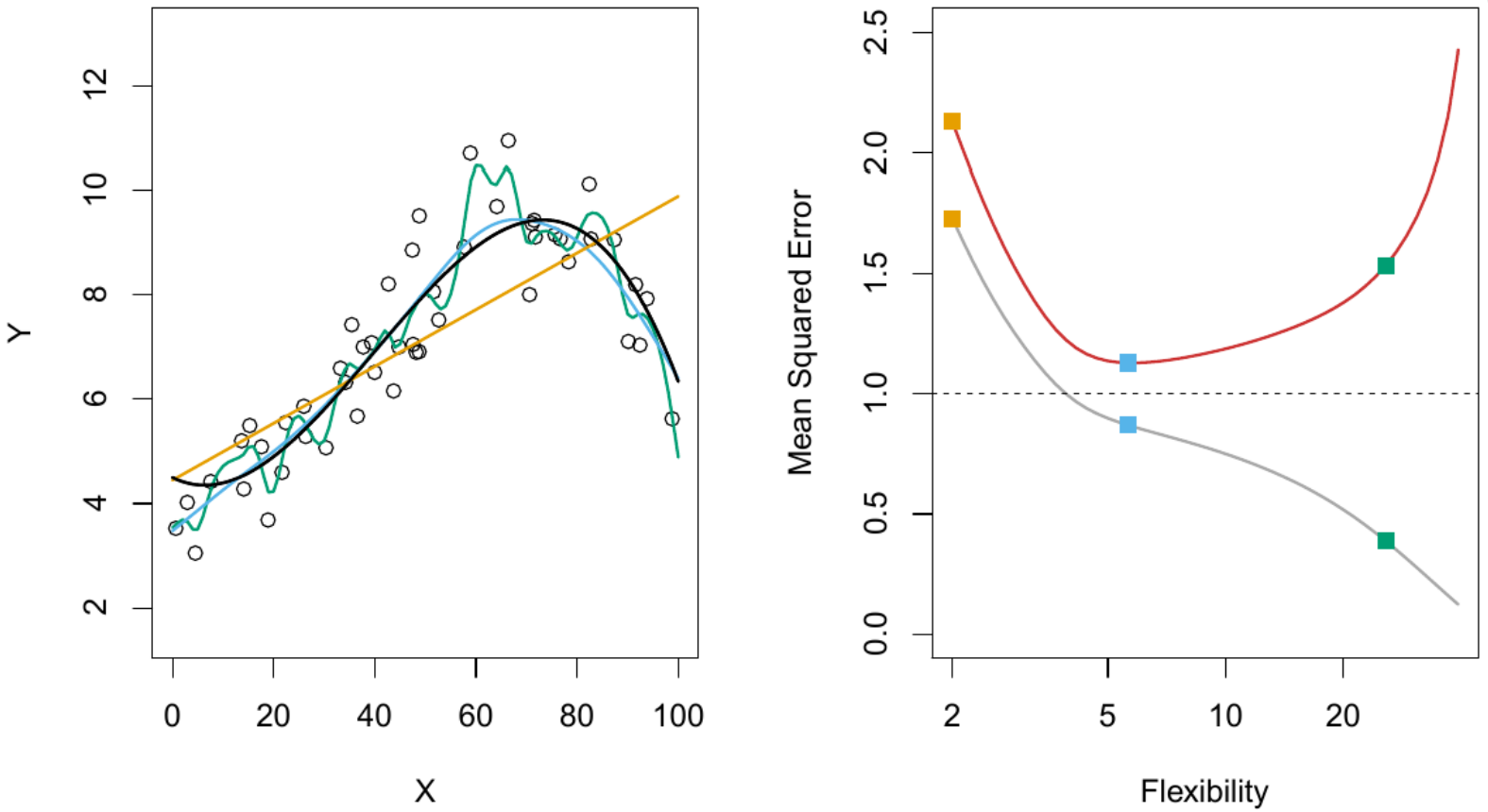
- training MSE (mean squared error):

$$\frac{1}{n}\sum_{i\in[n]}\left[y_i-\hat{f}(x_i)\right]^2.$$

- test MSE (ex: stock price tomorrow): unseen data pairs (x_0, y_0),

$$\text{Ave}_{(x_0,y_0)}[y_0-\hat{f}(x_0)]^2.$$

Minimizing the training MSE: decreasing/U-shape



models of different flexibility [r.h.s.: training/test MSE vs flexibility]

- low: linear regression (orange),
- medium: smoothing spline-1 (blue),
- high: smoothing spline-2 with increased flexibility (green).

$\Rightarrow$ high flexibility: danger of **overfitting**.

Bias-variance tradeoff: an interpretation of the U-shape

- Additive model: $Y = f(X) + \epsilon$.
- Dataset: $D = \{(x_i, y_i)\}_{i=1}^n$.
- Expected error on an unseen x_0 [derivation for the curious-minded]:

$$\mathbb{E}_{D,\epsilon}\left[y_0 - \hat{f}(x_0; D)\right]^2 = \overbrace{\text{Bias}_D^2[\hat{f}(x_0; D)]}^{\geq 0} + \overbrace{\text{Var}_D[\hat{f}(x_0; D)]}^{\geq 0} + \text{Var}(\epsilon),$$

$$\text{Bias}_D[\hat{f}(x_0; D)] = \mathbb{E}_D[\hat{f}(x_0; D) - f(x_0)],$$

$$\text{Var}_D[\hat{f}(x_0; D)] = \mathbb{E}_D\left(\hat{f}(x_0; D) - \mathbb{E}_D[\hat{f}(x_0; D)]\right)^2.$$

Bias-variance tradeoff - continued

Intuition of

$$\mathrm{Bias}_D[\hat{f}(x_0; D)] = \mathbb{E}_D[\hat{f}(x_0; D) - f(x_0)],$$
$$\mathrm{Var}_D[\hat{f}(x_0; D)] = \mathbb{E}_D\left(\hat{f}(x_0; D) - \mathbb{E}_D[\hat{f}(x_0; D)]\right)^2,$$
$$\mathrm{Var}(\epsilon),$$

- Var_D: sensitivity to D; high $\mathrm{Var} \approx$ flexible models $\Rightarrow$ overfitting.
- Bias_D: high $\mathrm{Bias} \approx$ can't capture complex data $\Rightarrow$ underfitting.
- $\mathrm{Var}(\epsilon)$: irreducible error.

Classification

- One can observe a similar bias-variance tradeoff.
- Training error rate:

$$\frac{1}{n}\sum_{i\in[n]}\mathbb{I}(y_i \neq \hat{y}_i).$$

- Test error rate: unseen data pairs (x_0, y_0),

$$\text{Ave}_{(x_0,y_0)}(\mathbb{I}(y_0 \neq \hat{y}_0))$$

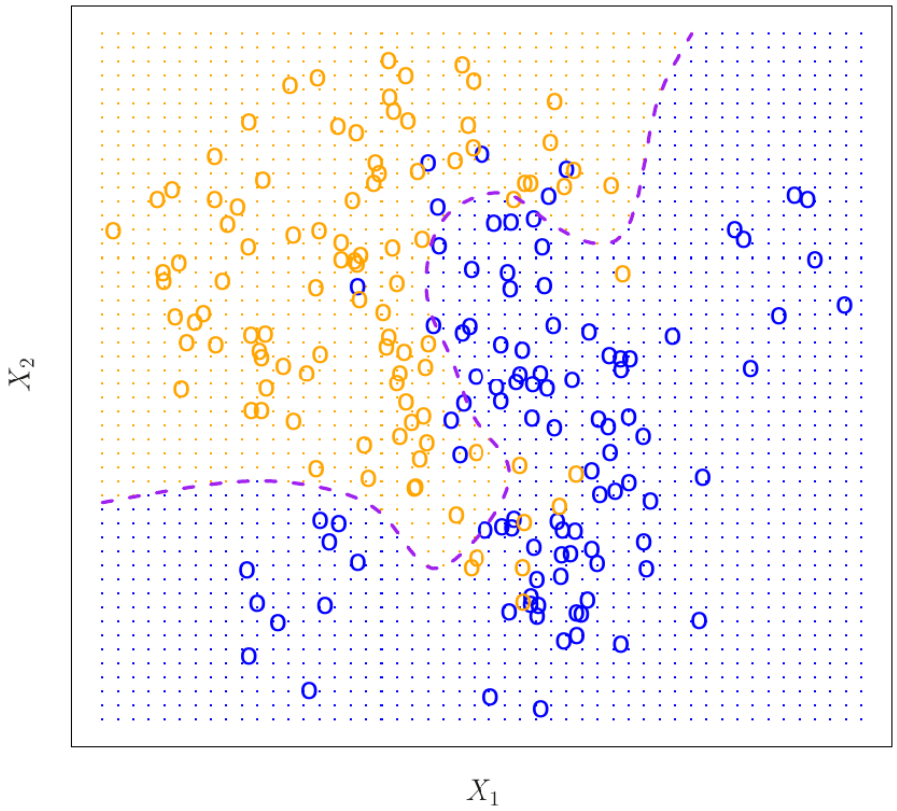
A 'good' classifier has small test error rate.

Bayes classifier

- The Bayes classifier minimizes the test error rate (Bayes error rate), and simple; for given input $\mathbf{x}_0$:

$$j^* := \arg\max_{j \in J} P(Y = j \mid X = \mathbf{x}_0).$$

- Spec ($|J| = 2$): predict class one if $P(Y = 1 \mid X = \mathbf{x}_0) > 0.5$.



An approximation of the Bayes classifier

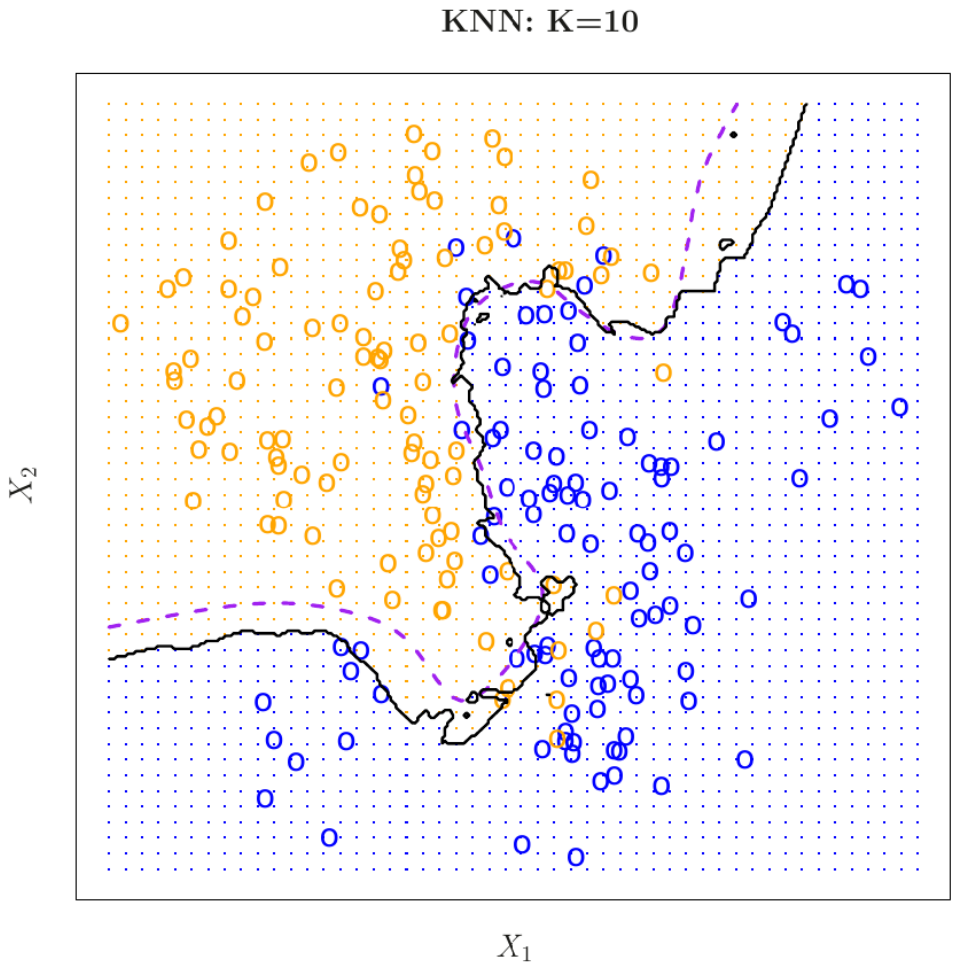
- Minor issue: $P(Y|X)$ is not available. $\Rightarrow$
- One could try to estimate $P(Y|X = \mathbf{x}_0)$; ex: KNN classifier.
- Given: $K \in \mathbb{Z}^+$. $\mathcal{N}_0 := K$ closest points to $\mathbf{x}_0$ in D. For fixed j:

$$\hat{P}(Y = j \,|\, X = \mathbf{x}_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} \mathbb{I}(y_i = j),$$

$$j^* := \arg\max_{j \in J} \hat{P}(Y = j \,|\, X = \mathbf{x}_0).$$

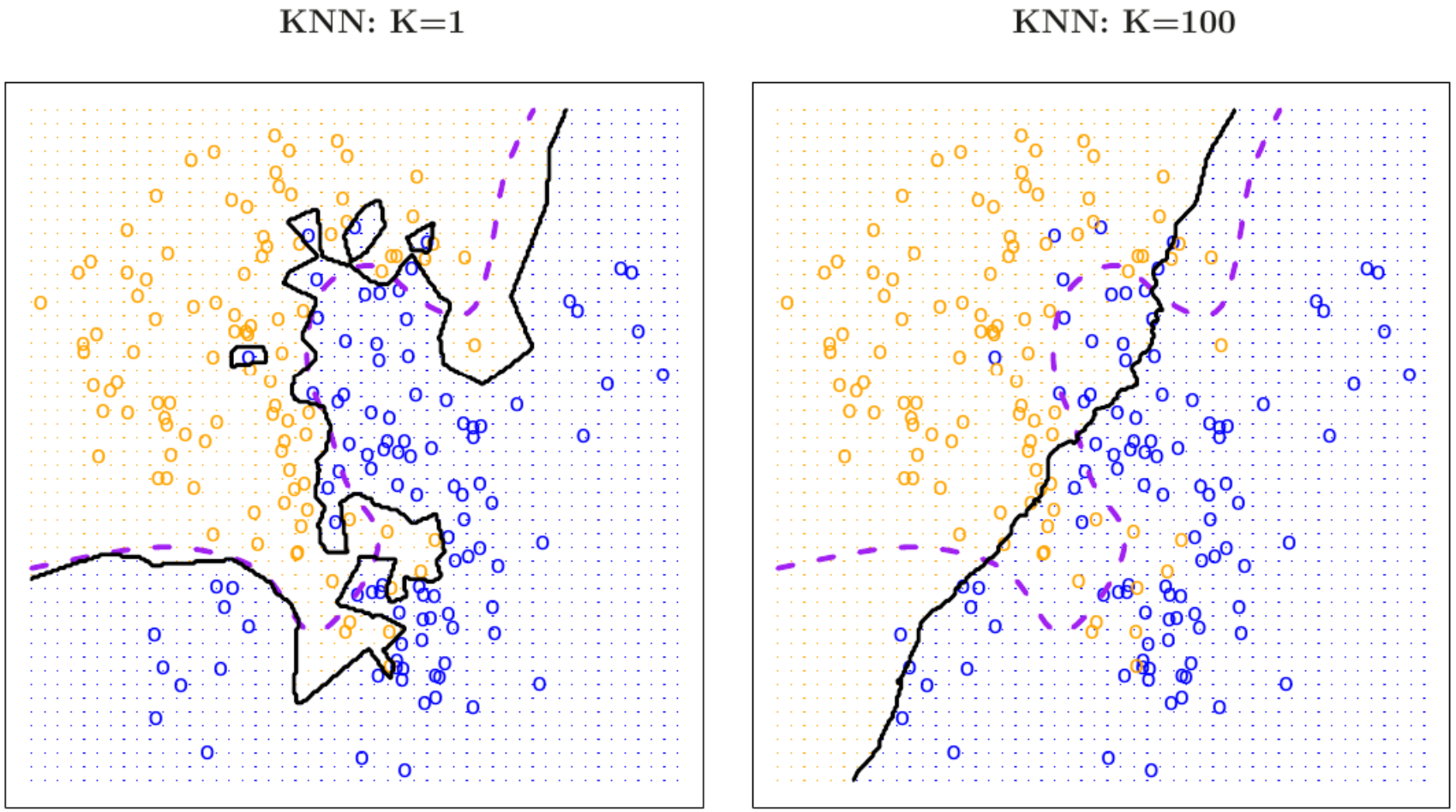
Idea: average the votes of kNNs, followed by the winning class j^*

KNN classifier: binary classification ($|J| = 2$), $p = 2$



- Bayesian decision boundary: purple,
- KNN decision boundary: black.

KNN classifier: binary classification, different K-s



- $K = 1$: too flexible decision boundary (low bias, high variance).
- $K = 100$: less flexible.

$\Rightarrow$ The choice of K can significantly affect the performance.

Broader context

- $(X, Y) \sim P \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$. Training data: $\{(x_i, y_i)\}_{i \in [n]} \overset{\text{i.i.d.}}{\sim} P$.
- Hypothesis class: $\mathcal{F} \subset \mathcal{Y}^{\mathcal{X}}$.
- Expected risk ('test error') with loss ℓ of a hypothesis $f \in \mathcal{F}$:

$$R(f) = \mathbb{E}[\ell(f(X), Y)] = \int_{\mathcal{X} \times \mathcal{Y}} \ell(f(x), y) \mathrm{d}P(x, y).$$

- Empirical risk ('training error') of an $f \in \mathcal{F}$:

$$R_n(f) = \int_{\mathcal{X} \times \mathcal{Y}} \ell(f(x), y) \mathrm{d}P_n(x, y) = \frac{1}{n} \sum_{i \in [n]} \ell(f(x_i), y_i).$$

- Bayes error: $R^* := \inf_{h: \mathcal{X} \to \mathcal{Y}, \text{ measurable}} R(f)$.
- Bayes hypothesis: $f \in \mathcal{F}$ if $R(f) = R^*$.
- Example losses: (i) squared loss: $\ell(x, y) = (x - y)^2$, (ii) 0-1 loss: $\ell(x, y) = \mathbb{I}(x \neq y)$.

We see quantities of the form:

$$\int_{\mathcal{X}} f(x)\mathrm{d}\mu(x).$$

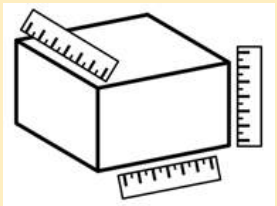
Question

What do they mean?

Idea

We define

1. a notion of length/area/volume/...: called measure (μ), then



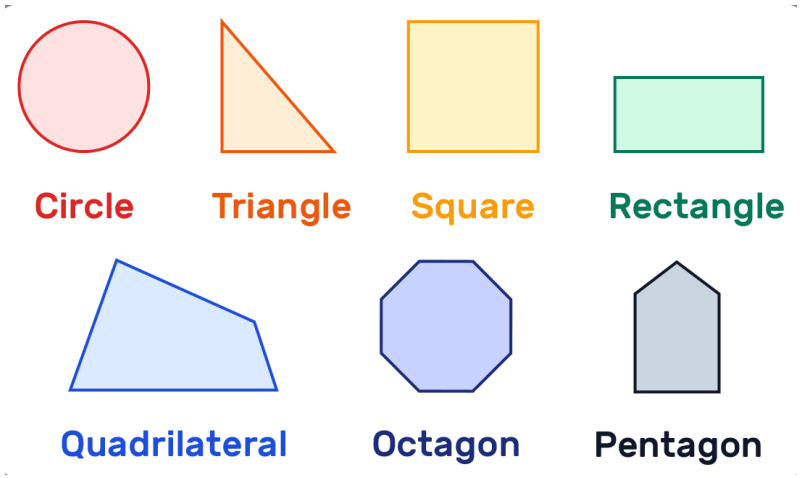
2. integrate with respect to it.

The space: $\mathcal{X}$

- Given: a set $\mathcal{X}$; say $\mathcal{X} = \mathbb{R}^d$.

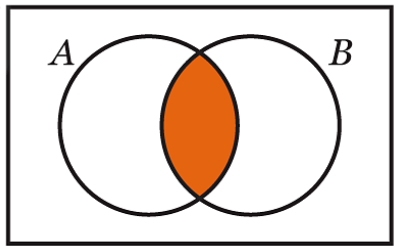
Question

- What are the subsets of $\mathcal{X}$ which have area?
- We call
  - them **measurable sets**: $\mathcal{A} \subseteq \mathcal{P}(\mathcal{X}) :=$ power set of $\mathcal{X}$,
  - $(\mathcal{X}, \mathcal{A})$ pair: **measurable space**.



Towards measurable sets: $\mathcal{A}$

It would be nice to be closed w.r.t. certain set operations, say



$A \cap B$

1. $A, B \in \mathcal{A} \Rightarrow A \cap B,\ A \cup B,\ A^c := \mathcal{X} \setminus A,\ A \setminus B,\ B \setminus A,\ A \triangle B \in \mathcal{A}$.
2. Finite operations: $A_1, \ldots, A_n \in \mathcal{A} \Rightarrow \cap_{i=1}^n A_i \in \mathcal{A},\ \cup_{i=1}^n A_i$.
3. (Countable) infinite operations: $A_1, A_2, A_3, \ldots \in \mathcal{A} \Rightarrow \cap_{i=1}^\infty A_i \in \mathcal{A},\ \cup_{i=1}^\infty A_i$.

Question

This looks cumbersome. How do we guarantee all these properties?

Measurable sets: $\mathcal{A}$

$\mathcal{A} \subseteq \mathcal{P}(\mathcal{X})$ is called **σ-algebra** (on $\mathcal{X}$; a.k.a. **σ-field**) if:

1. $\emptyset \in \mathcal{A}$,
2. $H \in \mathcal{A} \Rightarrow H^c \in \mathcal{A}$,
3. $H_n \in \mathcal{A}$ $(n \in \mathbb{Z}^+) \Rightarrow \cup_{n=1}^{\infty} H_n \in \mathcal{A}$.

Note

This small set of requirements guarantees all the closedness properties wanted (previous slide).

(Dummy) examples

$\mathcal{A} = \{\emptyset, \mathcal{X}\}$, $\mathcal{A} = \mathcal{P}(\mathcal{X})$.

Question

This definition seems slightly abstract. How do we come up with σ-algebras (beyond the 'dummy' ones)?

Construction of σ-algebra: generated σ-algebra

- Idea: we
  1. give certain sets we want to be measurable: $\mathcal{S} \subseteq \mathcal{P}(\mathcal{X})$.
  2. take the smallest σ-algebra containing it (see closure systems @ algebra).
- Formally, for $\mathcal{S} \subseteq \mathcal{P}(\mathcal{X})$, one considers

$$\sigma(\mathcal{S}) := \cap\{\mathcal{B} : \mathcal{S} \subseteq \mathcal{B}, \mathcal{B} \text{ is } \sigma\text{-algebra}\}.$$

Any σ-algebra celebrity?

- The Borel σ-algebra is among the most widely-used ones.

Topological space: $(\mathcal{X}, \tau)$

- Idea: $A \subseteq \mathbb{R}^d$ is open if all $x \in A$ can be surrounded by a ball in A.
- $\tau \subseteq \mathcal{P}(\mathcal{X})$ is a **topology** (on $\mathcal{X}$; a.k.a. **open sets**) if
  1. $\emptyset, \mathcal{X} \in \tau$,
  2. closedness w.r.t. any union: $A_i \in \tau$ $(i \in I) \Rightarrow \cup_{i \in I} A_i \in \tau$.
  3. closedness w.r.t. finite intersection: $A_i \in \tau$ $(i \in [n]) \Rightarrow \cap_{i \in [n]} A_i \in \tau$.

Borel σ-field

- Def: $\mathcal{B}(\mathcal{X}) := \sigma(\tau)$.
- $\mathcal{B}\left(\mathbb{R}^d\right) = \sigma(I)$, $I = \left\{ \times_{i=1}^{d} [a_i, b_i], a_i, b_i \in \mathbb{R}, i \in [d] \right\}$ intervals.

But what is a measure?

Measure, probability measure

- Given: $(\mathcal{X}, \mathcal{A})$ measurable space.
- $\mu : \mathcal{A} \to \bar{\mathbb{R}} := \mathbb{R} \cup \{\pm\infty\}$ is called **measure** if it is
  1. normalized: $\mu(\emptyset) = 0$,
  2. non-negative: $\mu(H) \geq 0$ for all $H \in \mathcal{A}$ (this can be relaxed), and
  3. σ-additive: for $H_1, H_2, \ldots \in \mathcal{A}$ pairwise disjoint

$$\mu\left(\cup_{n=1}^{\infty} H_n\right) = \sum_{n=1}^{\infty} \mu(H_n).$$

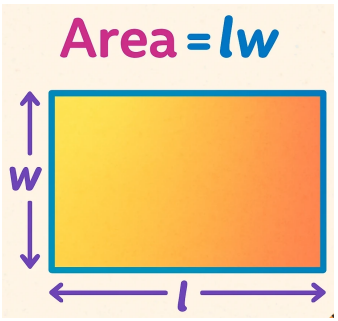
Notes:

- The triplet $(\mathcal{X}, \mathcal{A}, \mu)$ is called **measure space**.
- A measure μ is called **probability measure** if $\mu(\mathcal{X}) = 1$.

Integration of non-negative simple functions; measurable functions

- Let $f = \sum_{i=1}^{n} c_i \chi_{A_i}$, where $c_i \in \mathbb{R}_{\geq 0}$, $A_i \in \mathcal{A}$, $i \in [n]$ (w.l.o.g. pairwise disjoint). Then the integral of this **simple function** is

$$\int_{\mathcal{X}} f(x) \mathrm{d}\mu(x) := \sum_{i=1}^{n} c_i \mu(A_i).$$



- An $f : (\mathcal{X}, \mathcal{A}) \to (\mathcal{Y}, \mathcal{B})$ is called **measurable** if

$$B \in \mathcal{B} \Rightarrow f^{-1}(B) \in \mathcal{A}.$$

Example:

- $(\mathcal{Y}, \mathcal{B}) = \left(\mathbb{R}^d, \sigma\left(\tau_{\mathbb{R}^d}\right)\right)$; for us $d = 1$ will the job.

Integration of non-negative functions, towards general ones

Let $(\mathcal{X}, \mathcal{A}, \mu)$ be a measure space.

- For $f : (\mathcal{X}, \mathcal{A}) \to \mathbb{R}_{\geq 0}$ measurable, let

$$\int_{\mathcal{X}} f(x) \mathrm{d}\mu(x) := \sup \left\{ \int_{\mathcal{X}} g(x) \mathrm{d}\mu(x) \, : \, 0 \leq g \leq f; g \text{ is simple} \right\}.$$

- For $f : (\mathcal{X}, \mathcal{A}) \to \mathbb{R}$ measurable, let
  - $f^+ := \max(f, 0)$ be its positive part.
  - $f^- := -\min(f, 0)$ be its negative part; $f = f^+ - f^-$.

Integration of general functions

If

- at least one of $\int_{\mathcal{X}} f^+(x)\mathrm{d}\mu(x)$ and $\int_{\mathcal{X}} f^-(x)\mathrm{d}\mu(x)$ is finite,
- then (the expression below makes sense and) let

$$\int_{\mathcal{X}} f(x)\mathrm{d}\mu(x) := \int_{\mathcal{X}} f^+(x)\mathrm{d}\mu(x) - \int_{\mathcal{X}} f^-(x)\mathrm{d}\mu(x).$$

- f is called **finitely integrable** if $\int_{\mathcal{X}} f(x)\mathrm{d}\mu(x) < \infty$.

This is the integral we referred to (expected risk, empirical risk).

Local summary

- tasks: prediction, inference,
- methods: parametric, nonparametric,
- model accuracy & flexibility: regression, classification,
- broader context: risk.

